evolution of virulence

19 October 2025

Definitions

- **virulence**
  - (broad): decrease in a host's fitness caused by a parasite.
  - (narrow): *per capita* rate of parasite-induced host mortality
- **resistance**: host's ability to resist or minimize infection
- **tolerance**: host's ability to support parasite infection without losing fitness
- **case mortality** (CM): fraction of hosts killed by infection
- Parasite-host interaction complicates the definition of virulence (assumes that a more virulent parasite is more virulent for *all* host genotypes/species)
- conceptually:
  - parasite load depends on balance between parasite *within-host reproduction rate* and host's *parasite clearance rate*
  - virulence depends on parasite load and per-copy parasite *pathogenicity* and host *tolerance*
  - this establishes the terms of the arms race, but these components can't be separated if we look at a single host-parasite pair (parasite virulence is often confounded with host tolerance)
- all in an arms race ("trench warfare", quantitative resistance) rather than RQ context (qualitative resistance)

Classical dogma

- Parasites evolve lower virulence over time "for the good of the species". Group-selectionist *but* some evidence? (Méthot 2012)
  - syphilis; first seen in Europe in 1495 (the "Great Pox") (Knell 2004)
    - origins? (previously misdiagnosed; evolved increased virulence; from Africa; from the New World)
    - virulence decreased rapidly over 50 years (maybe even 5-7 years?)
  - *virgin-soil epidemics*: smallpox, etc. (Crosby 1976; Ostler 2020) (probably *not* virulence: lack of genetic resistance, previous exposure, societal breakdown, effects of colonization?)
- sampling bias?
  - biocontrol efforts always select for maximal virulence (even though *intermediate* virulence may have more impact on host populations)
  - mild introductions may not be noticed

Tradeoff theory

- Intermediate virulence evolves due to *host-level* selection (group theory returns); a tradeoff between transmission *rate* (infections/host/time) and virulence (*defined as mortality/time*) maximizes R_0 (total

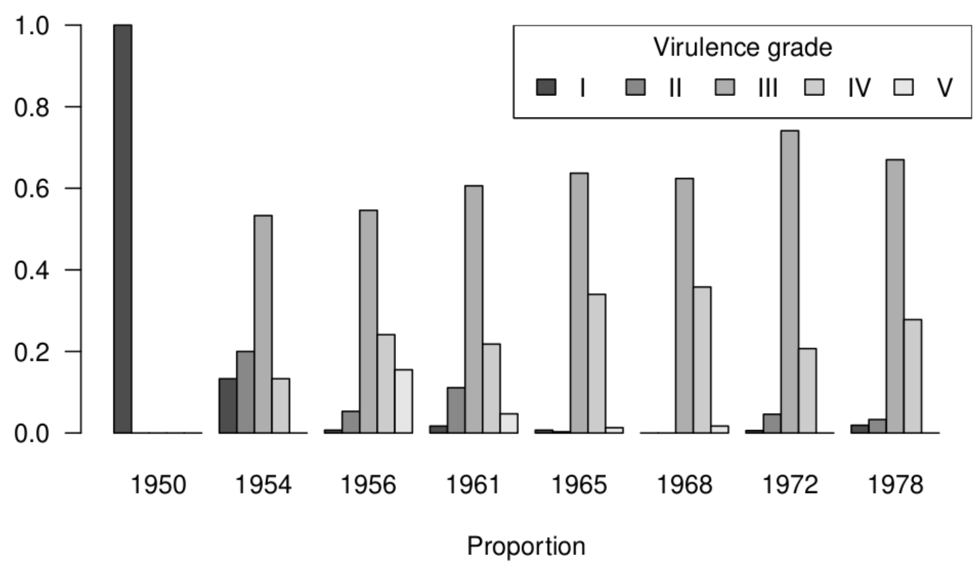
transmission per generation) at **intermediate** virulence.
- conceptually, mediated by parasite replication rate or load (cf spore production/*Daphnia* fecundity example (Decaestecker et al. 2007)

Example: myxomatosis

Viral disease; mild in Brazilian rabbits (*Sylvilagus brasiliensis*), virulent in European rabbits (*Oryctolagus cuniculus*). Mosquito- and flea-borne. Introduced (several times) in Australia to control introduced rabbits, finally spread 1950-1951. Case mortality originally >99%, populations initially decreased by 90%. CM initially dropped to 90%, then further. Resistance: test by infecting laboratory rabbits that haven't evolved. CM of grade III strain drops from 90% to about 50% as populations experience more epizootics. At the same time mean virus grade drops from I to III, then rebounds.

Evidence for tradeoff theory: Higher grades (higher case mortality) also have faster mortality (<13 days to >50 day survival as CM goes from >99% to <50%). Skin virus *titer* is also higher (and increases faster with time) for higher grades. Mosquito infection probability is proportional to skin titer. (Some biological complications.)

Myxomatosis grades over time (Fenner et al., 1956)



Bottom line: myxomavirus probably still reduces populations somewhat, but the Australians continue to look for other biocontrol solutions (calicivirus, rabbit haemhorrhagic disease).

Lab experiments on titers, transmission probabilities, etc. etc. etc. (Fenner, Day, and Woodroofe 1956); simulation model (Dwyer, Levin, and Buttel 1990)

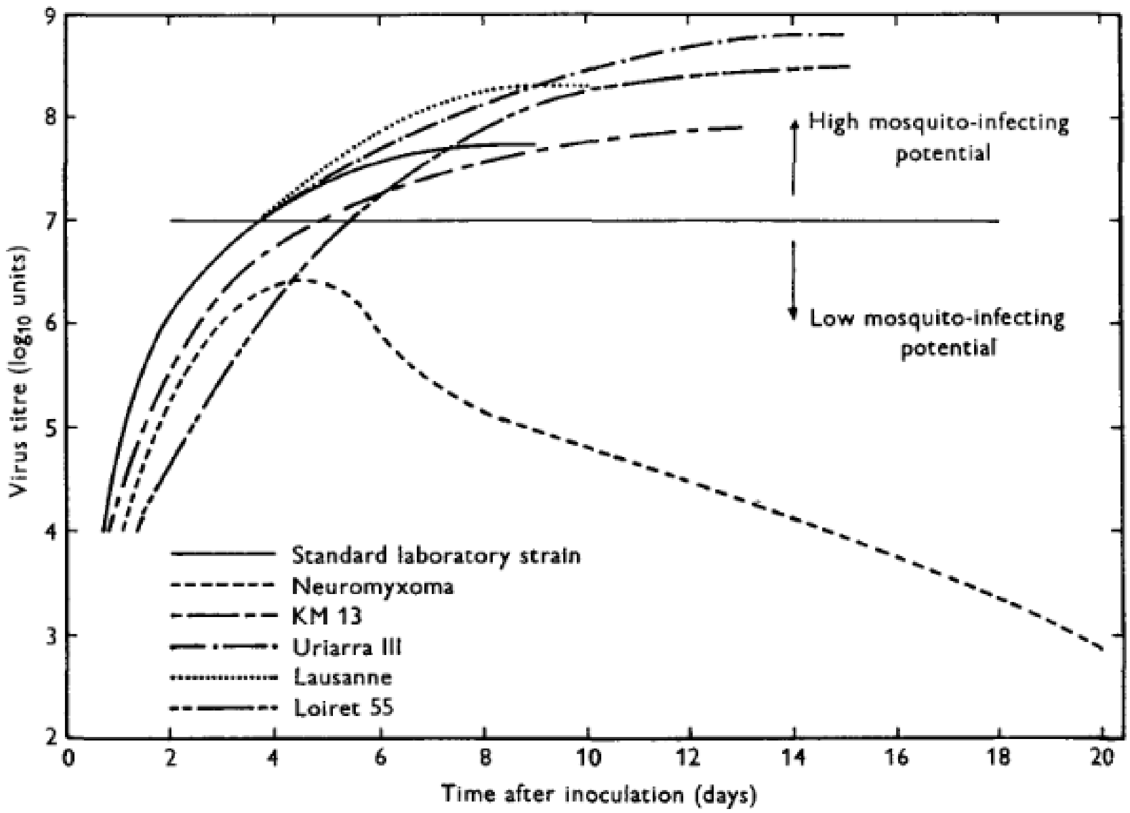


Fig. 1. The titre of virus in slices of skin taken from the surface of lesions produced by the intradermal inoculation of rabbits with large doses ($10^{4\cdot 3}$ rabbit-infectious doses) of various strains of myxoma virus.

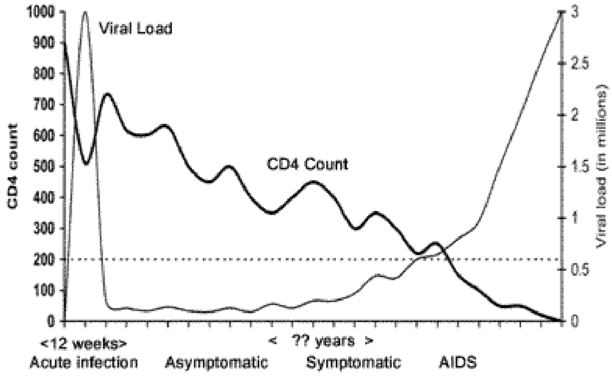
Genomic analysis: Kerr et al. (2012), Kerr et al. (2013), Kerr et al. (2022)

- Australia
  - some mutations with clear virulence effects (insertion disrupts reading frame involving cell cycle; deletion affects immunosuppressive pathway)
  - probably "attenuation-restoration": new mutations reverse previous evolutionary attenuation > Whether MYXV and European rabbits can coevolve [to tolerance/lower virulence] probably depends on whether systemic spread and multiple cutaneous lesions leads to more overall transmission and better fitness than does a single fibroma-like lesion and on whether European rabbit populations have immune system alleles that could constrain virus replication to the initial cutaneous site. If dissemination is critical for transmission, then mutations in the virus that favor dissemination by suppressing immune responses may also increase virulence.
- Britain
  - premature stop codon disrupts immunosuppression
  - parallel evolutions, but different substitutions

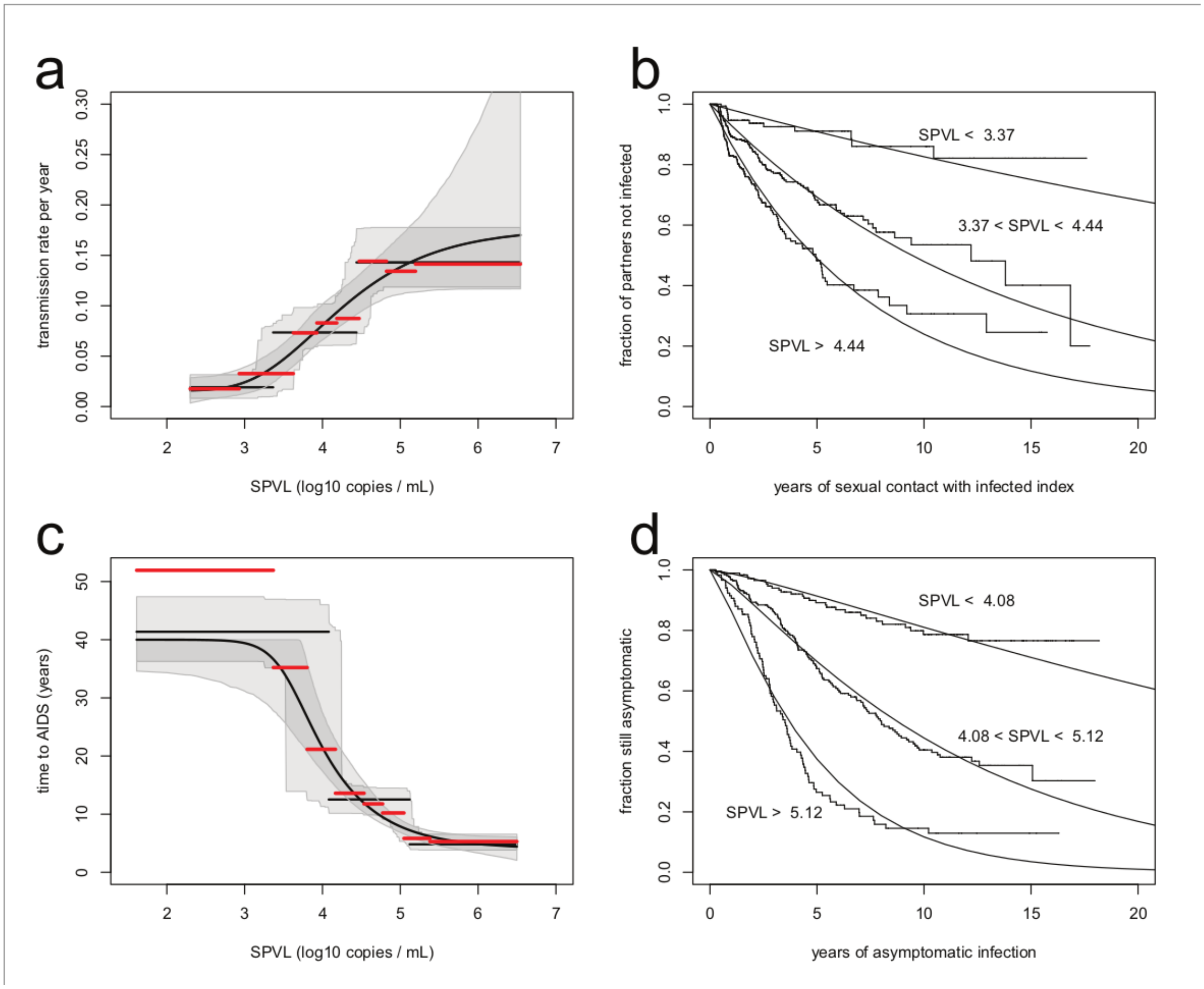
Example: HIV

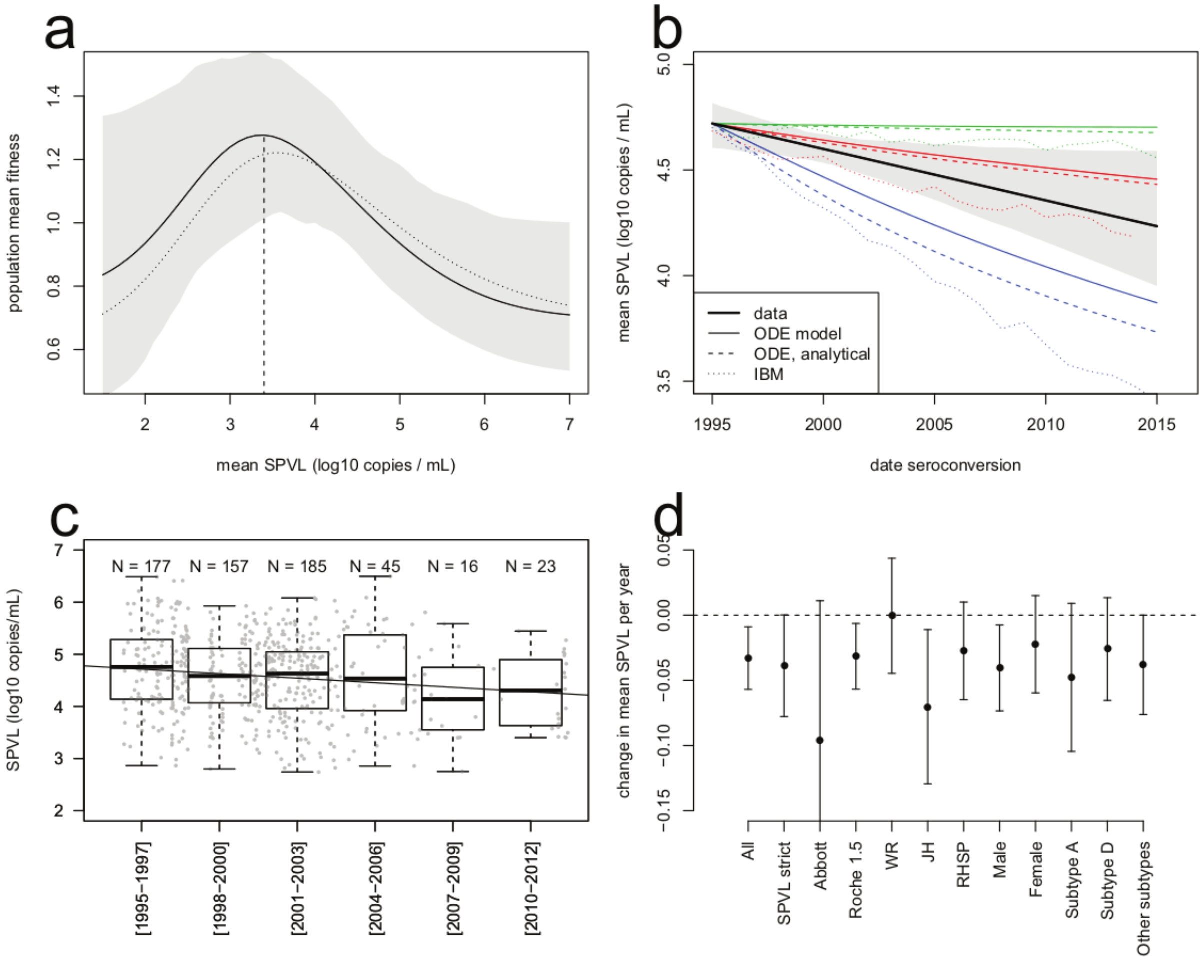
- Correlations among
  - *setpoint viral load*
  - *time to progression* or *rate of CD4 decline* (mechanisms still poorly understood! within-host evolution for diversity, virulence, immune escape? immune aging?? accumulation of opportunistic infections?)
  - *transmission probability* (as measured in *serodiscordant couples*; *Rakai cohort*)

- no longer ethically measurable

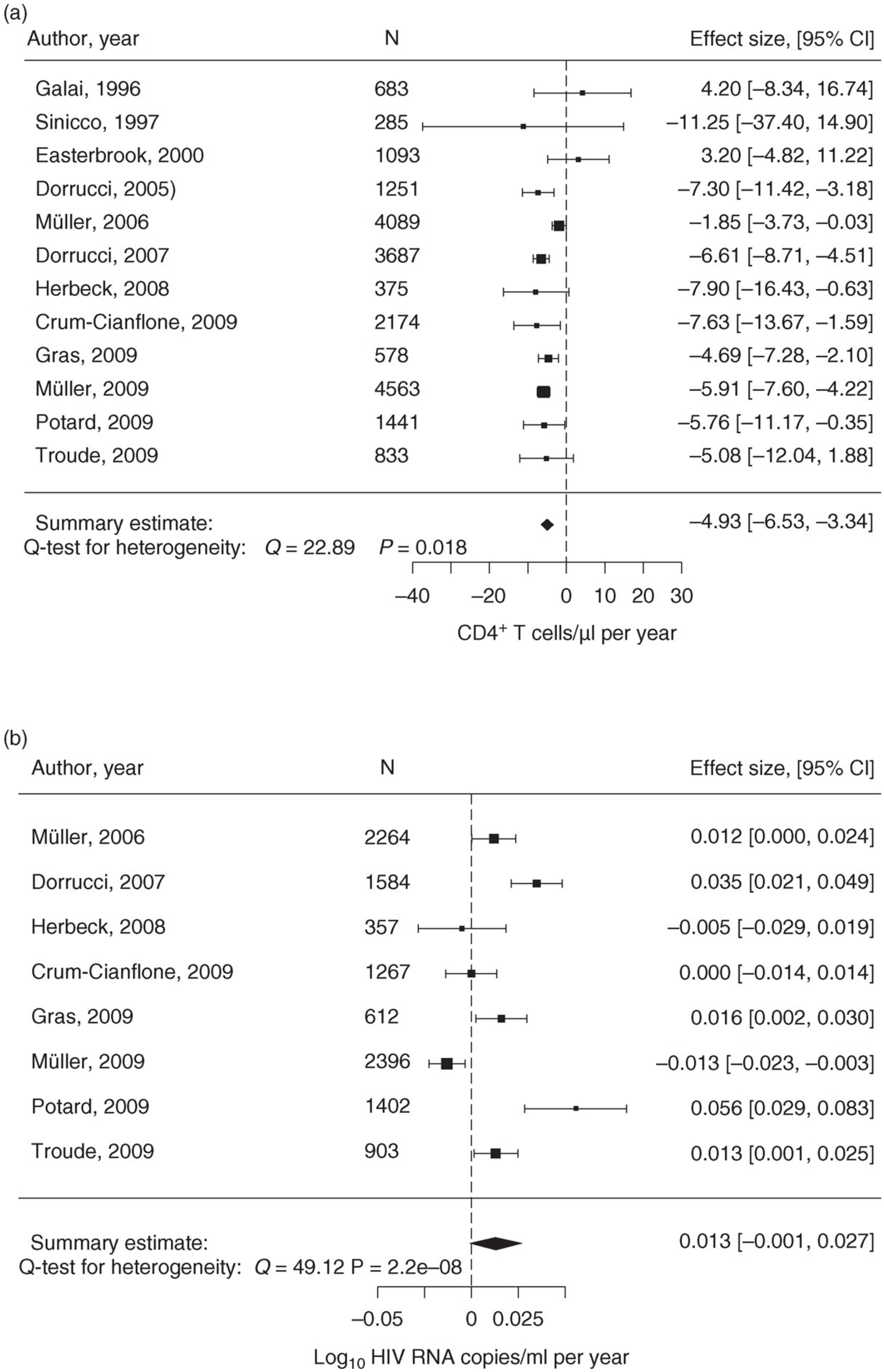


https://www.thebodypro.com/article/course-hiv-disease





- some suggestion of overall increase in virulence (decreased CD4 count/increased viral load)
- highly variable (e.g. increasing in Italy (Müller et al. 2009)? attenuating due to spread of less virulent subtype C (Ariën, Vanham, and Arts 2007)? decreasing in Uganda (Blanquart et al. 2016)? increasing overall (Herbeck et al. 2012)?

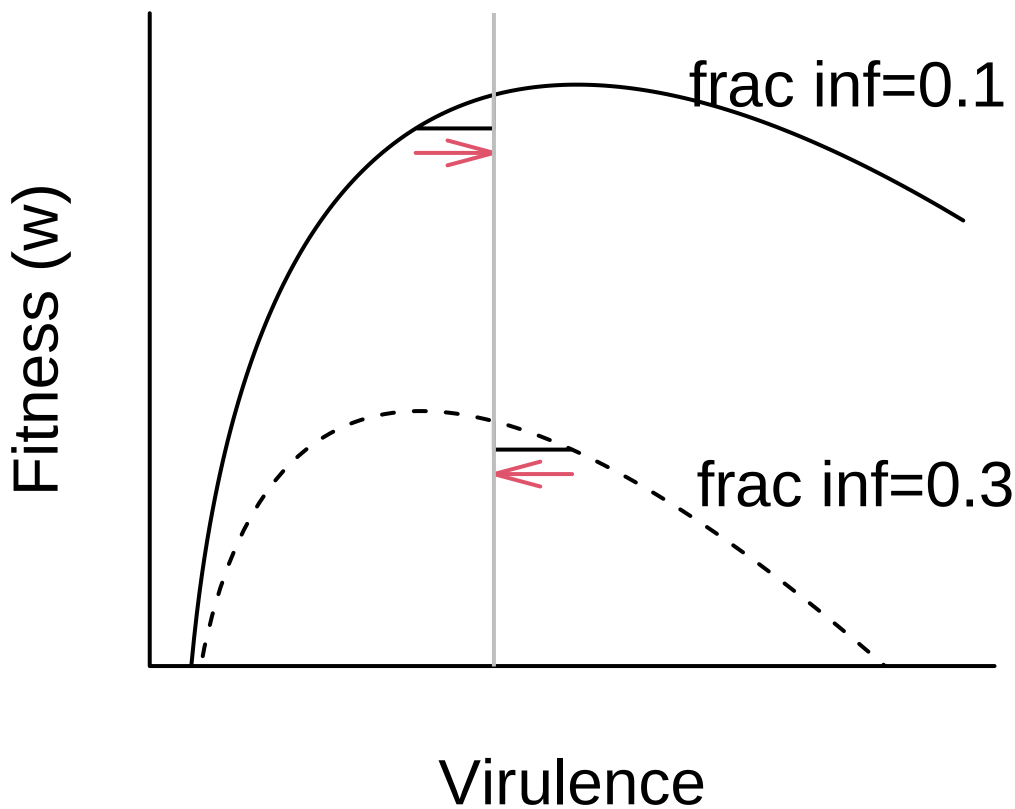

(a)
Author, year
N
Effect size, [95% CI]
Galai, 1996
683
4.20 [–8.34, 16.74]
Sinicco, 1997
285
–11.25 [–37.40, 14.90]
Easterbrook, 2000
1093
3.20 [–4.82, 11.22]
Dorrucci, 2005)
1251
–7.30 [–11.42, –3.18]
Müller, 2006
4089
–1.85 [–3.73, –0.03]
Dorrucci, 2007
3687
–6.61 [–8.71, –4.51]
Herbeck, 2008
375
–7.90 [–16.43, –0.63]
Crum-Cianflone, 2009
2174
–7.63 [–13.67, –1.59]
Gras, 2009
578
–4.69 [–7.28, –2.10]
Müller, 2009
4563
–5.91 [–7.60, –4.22]
Potard, 2009
1441
–5.76 [–11.17, –0.35]
Troude, 2009
833
–5.08 [–12.04, 1.88]
Summary estimate:
–4.93 [–6.53, –3.34]
Q-test for heterogeneity: Q = 22.89 P = 0.018
–40
–20
0
10
20
30
CD4+ T cells/μl per year
(b)
Author, year
N
Effect size, [95% CI]
Müller, 2006
2264
0.012 [0.000, 0.024]
Dorrucci, 2007
1584
0.035 [0.021, 0.049]
Herbeck, 2008
357
–0.005 [–0.029, 0.019]
Crum-Cianflone, 2009
1267
0.000 [–0.014, 0.014]
Gras, 2009
612
0.016 [0.002, 0.030]
Müller, 2009
2396
–0.013 [–0.023, –0.003]
Potard, 2009
1402
0.056 [0.029, 0.083]
Troude, 2009
903
0.013 [0.001, 0.025]
Summary estimate:
0.013 [–0.001, 0.027]
Q-test for heterogeneity: Q = 49.12 P = 2.2e–08
–0.05
0
0.025
Log10 HIV RNA copies/ml per year

Example: SARS-COV-2

- successive waves tended to have higher $\mathcal{R}_0$
- higher $\mathcal{R}_0$ partly from *immune evasion*; re-infection of individuals recovered from/immune to previous strains
- virulence changes: alpha, delta higher, omega lower (reduced "fusogenicity" [ability to fuse with/enter host cells]; tropism for cells higher in respiratory tract? (Carabelli et al. 2023)

Theory

- if there is a tradeoff, we would expect expect strong effects of **transmission mode**
  - vector-borne > direct
  - high virulence for "necrotransmission" (via dead hosts: anthrax, chronic wasting disease)
  - horizontal transmission > vertical
  - needle-borne > STD?
  - environmental (water-borne, e.g. cholera) > direct
- does higher overall transmission rate (due to population density, poor hygiene, etc.) select for higher transmission?
- **facultative** parasites (e.g. soil-borne microbes with a **facultative** stage) should be more virulent
- "curse of the pharaoh": effect of resting stages? (Bonhoeffer, Lenski, and Ebert 1996)
- spatial restriction should? decrease virulence (Kamo and Boots 2006)
- Maximizing R_0:



- "Virulence" could be effect of host mortality, or rapid clearance.

Within-host competition

- basic tradeoff theory assumes one infection/strain per host
- effects of mutation, **superinfection**: within-host competition
- tends to *increase* optimal virulence

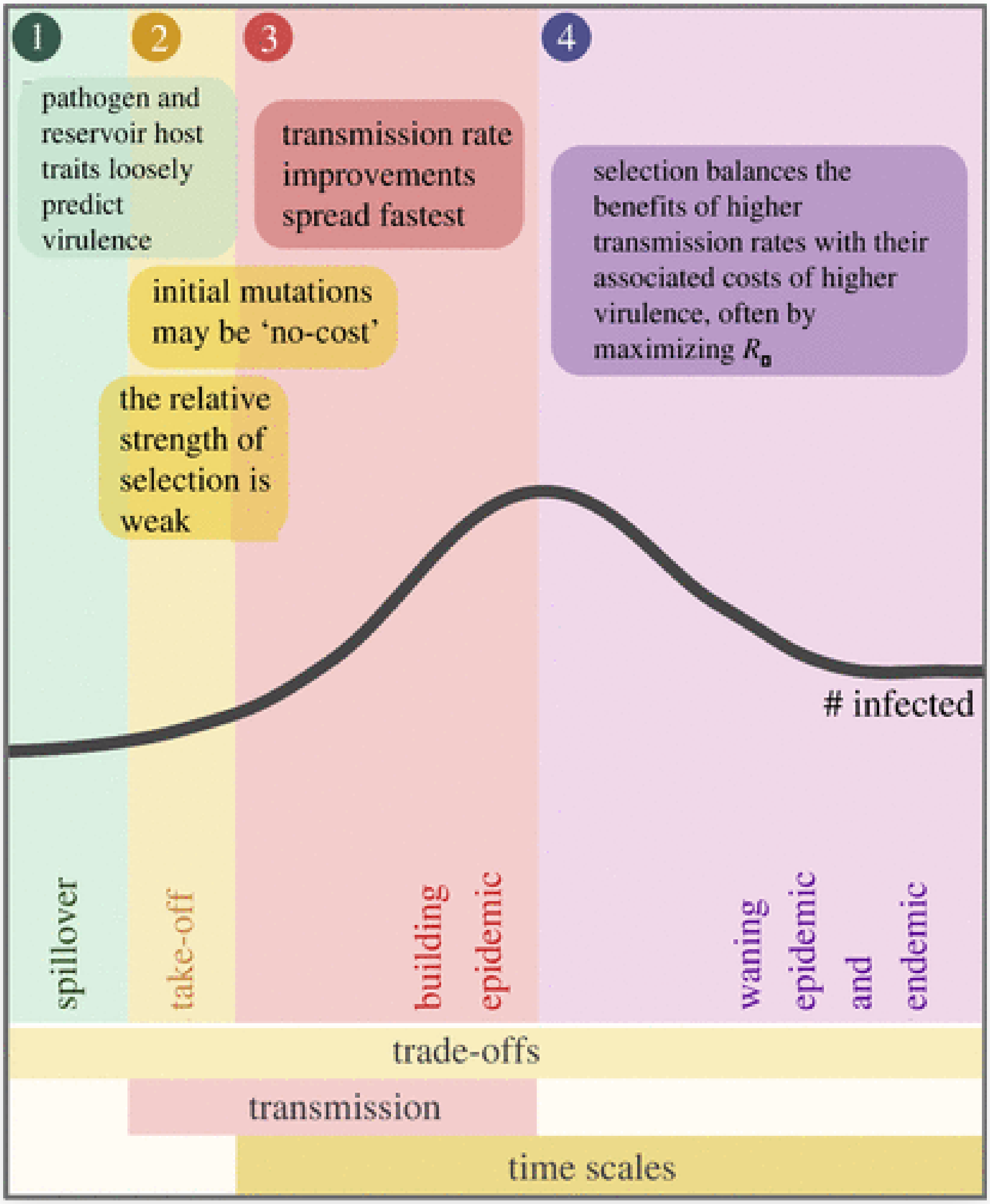
Short-sighted evolution

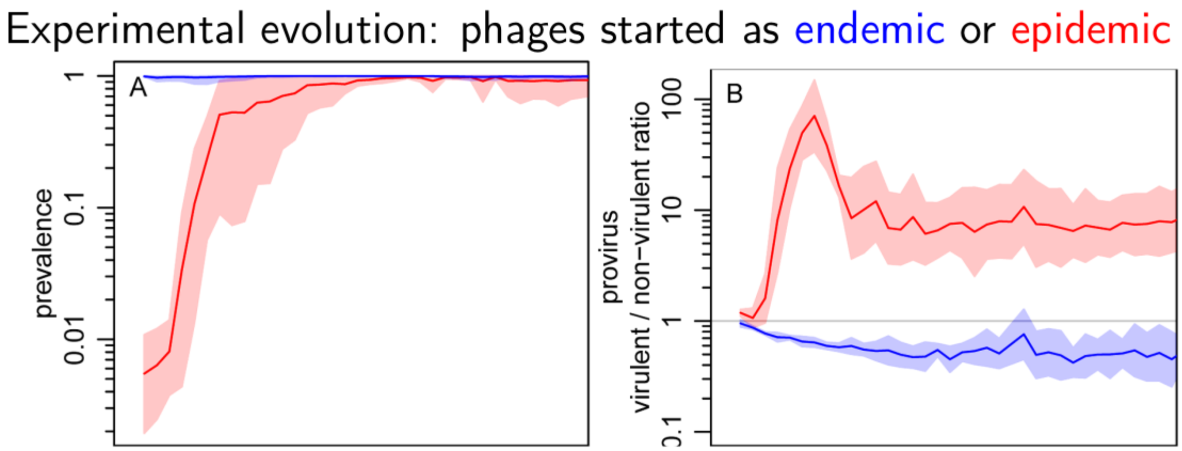
- sometimes evolution is just stupid (Levin and Bull 1994)
- meningitis-producing, paralytic polio strains (central nervous system tropism)
- HIV [most transmission probably occurs during acute phases]

Epidemic vs. endemic phases; transient virulence

- Most theory assumes that disease is at an *endemic equilibrium*, so that *lifetime fitness* (i.e. R_0 maximization) is what matters
  - this also means that increasing overall transmission (due to population density, poor hygiene, etc.) **doesn't** select for higher virulence
- During the exponential growth phase of an epidemic, *speed of increase* (r maximization) is what matters
  - optimal virulence is higher than for endemic equilibrium
- We expect *transient* selection for higher virulence at the beginning of an epidemic

(Frank 1996; Bolker, Nanda, and Shah 2010; Visher et al. 2021; Day and Proulx 2004; Berngruber et al. 2013; Park and Bolker 2017)


1
2
3
4
pathogen and reservoir host traits loosely predict virulence
transmission rate improvements spread fastest
selection balances the benefits of higher transmission rates with their associated costs of higher virulence, often by maximizing R_0
initial mutations may be ‘no-cost’
the relative strength of selection is weak
infected
spillover
take-off
building epidemic
waning epidemic and endemic
trade-offs
transmission
time scales



Effects of vaccines and treatment

- evolution due to **risk compensation** (Massad et al. 2006)?
- evolution of higher virulence in unvaccinated people due to "leaky" vaccination (Gandon et al. 2001)?
- mouse malaria: (Margaret J. Mackinnon and Read 2004; M. J. Mackinnon, Gandon, and Read 2008); consistent with "arms race" upregulation of replication
- increased virulence in Marek's disease: reduced host generation time or effects of leaky vaccine? (Atkins et al. 2013)
- in HIV due to antiretroviral therapy (Herbeck et al. 2016)?

M. J. Mackinnon, Gandon, and Read (2008):

> a cautionary approach to the widespread use of anti-replication or anti-disease vaccines seems justified. Ideally, this means combining such vaccines with transmission-blocking vaccines, bednets, drugs, housing improvements and other transmission-reducing measures

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Last updated: 2025-10-31 09:49:43.390203